

Section 1: Introduction

Chapter 2. Introduction to Physique Enhancement

Physique sport is a new phenomenon that has changed substantially in the last several decades. However, it has deeper roots that go back a century. These roots might surprise some competitors, potentially opening their eyes to new, healthier relationships with bodybuilding. Knowing that bodybuilding and strength training were once seen as the same, and that they were practiced for the pursuit of health, is an important perspective often absent from modern conversations. However, this is not to say that physique athletes or coaches should live in the past. Modern approaches to training which focus specifically on hypertrophy have contributed to the continual physique improvement of bodybuilders over time. However, coaches need to be aware of the differences and similarities between competitors and non-competitors, the latter of whom do not need to use as specialized of approaches. When dealing with competitors, coaches need to be versed in the competitive options in untested and natural federations, and the various categories physique athletes can compete in. Further, coaches and competitors alike need to be aware of the physiological and psychological stresses of the contest preparation process, the mental health risks associated with physique sport, and how to mitigate such risks while staying in one's scope of practice. Finally, coaches should understand the importance of evidence-based practice, ensuring they integrate not only their own expertise into their coaching, but also the experiences of their clients, and the best available scientific research.

Key Takeaways

- Physique sport was born out of physical culture, and has a rich history where training for strength, aesthetics, and health were one in the same.
- Modern approaches to bodybuilding are more specialized, and ultimately, more appropriate for the increased demands of modern bodybuilding, but they should still be sustainable.
- Contemporary physique athletes have multiple categories which they can compete in, which operate on a spectrum of how much muscularity is emphasized.
- Further, competitors have the option of competing in natural, drug-tested competitions, or in untested competitions. At higher levels of untested competition, competitors are much more muscular, but shoulder increased health risks and potential legal ramifications.
- However, not all physique athletes compete, and non-competitors may make up a large proportion of your clients. These non-competitive physique athletes will share some, but not all the same needs as competitors, as they would not go through contest preparation.

- The process of competition preparation is both mentally and physically arduous. It typically takes multiple months to get as lean as necessary to compete, and then multiple months to recover from this state, and the process of getting there.
- The contest preparation process is one factor which may contribute to higher rates of body image and eating disorders among physique competitors. Other factors are which more individuals with a history or propensity of developing eating or body image issues may be attracted to the sport, and many traditional methods used by bodybuilders are rigid, and restrictive.
- Using more flexible approaches to nutrition, and facilitating the transition between contest preparation, to recovery, and back to “normal” offseason eating behaviors is challenging but can help mitigate the mental health risks of competing.
- Coaches need to know the signs and symptoms of common disorders but must remember their scope of practice. Having a referral network which includes an appropriate mental health care practitioner is critical to safeguarding ones’ clientele.
- In all domains of training, be it nutrition, training, or coaching, an evidence-based framework should be applied to practice.
- Coaches should not rely exclusively on scientific data since there are not studies on all topics and group averages do not always apply to individuals.
- Likewise, coaches should not rely exclusively on experience, as anecdotal evidence can’t establish causation. While some traditional practices might be sufficient to get the job done, they might not be optimal, safe, sustainable, or necessary.
- Therefore, experience and science should be combined, playing complementary roles in coaching, which is integrated collaboratively with the client, considering their unique needs, preferences, experiences, and values.

Section 2: Physiology, Anatomy & Biomechanics of Physique Enhancement

Chapter 3. The Personal Training Profession Summary

Based on the multiplicity of factors that a Physique Coach must understand and implement with their physique athletes’ nutrition, training, and condition approach, an appreciable understanding of components is critical for the physique athletes’ safety and performance as they progress towards their goal of contest readiness. A foundational approach to this process is a valuable appreciation and understanding of human anatomy and physiology and its relation to neurology, endocrinology, kinesiology, biomechanics, and motor learning and control. An adequate understanding on this foundational information will allow for a much

more lucid understanding of other critical components of physique preparation, such as nutrition and micro- and macro-nutrient metabolism, just to name a few.

Key Takeaways

- Physique-related competing requires the understanding of the basics of anatomy and physiology and it is important that this understanding be incorporated into the training and conditioning program.
- The nervous system is critical in incorporating information about sensation and movement; therefore, a physique-related competitor and/or coach needs to understand the role the nervous system plays in training and conditioning.
- Nerves signal muscle to contract and cause joint movements; therefore, an understanding of the skeletal system, the types of joint, what movements are allowed at various joint, and how the skeletal system is impacted by physical activity can help influence the approach to training.
- The action of joints within the skeletal system are based on the principle of mechanical levers and provides the basis of locomotion.
- There are different types of muscle based on their shape; however, skeletal muscle is a highly dynamic tissue indicated by the manner in which muscle fiber types change and adapts with resistance training.
- Muscle contraction is a dynamic biochemical process influenced by the nervous system, enzymes, and electrolytes and inevitably occurs to produce muscle actions that are either concentric, eccentric, or isometric.
- Muscle contraction provides the basis for joint movement and mechanical work in the ability they possess to produce force, torque, and power.
- The endocrine system is associated with hormones regarding physique-related competing, important in regulating metabolic rate, energy expenditure, and metabolism, as well as muscle protein synthesis and hypertrophy.

Chapter 4. Essentials of Metabolism, Energy Balance, and Fat Loss

At its core, human metabolism is the process of converting chemical energy found in food into different forms of energy the body needs to function: Movement (kinetic energy), heat, and other forms of chemical energy. Over millennia, the human body has developed multiple different sets of chemical reactions to create the chemical energy needed for the body to function. These different sets of chemical reactions utilize a wide range of substrates that the body obtains from food eaten to produce enough energy. These energy systems are utilized differentially based on the needs of the body at any given time. Specific to the Physique Coach, there are several core concepts to consider when working with physique clients.

Key Takeaways

- Exercise intensity and duration primarily determine whether the body utilizes more anaerobic metabolism or aerobic metabolism, which determines whether the body is primarily utilizing carbohydrates or fatty acids to fuel exercise.
- Energy balance is a critical concept for coaching physique clients, and clients will need to be in energy surpluses and energy deficits during specific periods of their training cycles.

- How metabolisms adapt during periods of both deficits and surpluses is critical to understanding what aspects of energy expenditure clients need to focus on in order to optimize their results.
- One of the key considerations the Physique Coach should consider when working with physique clients is to be aware of low-energy availability and the potential health consequences that can arise from being in that state for too long.
- Fat tissue is not simply an inert tissue, it is a metabolically active organ that plays a critical role in overall human metabolism and human health. The Physique Coach should understand how subcutaneous fat and visceral fat have different impacts on health and how different types of fat cells serve different functions in the body. Additionally, exactly how fat loss occurs is a core concept for the Physique Coach and their clients.
- Both diet and exercise are critical for weight loss for the physique athlete. However, they have very different effects on the body. Diet is the primary tool for creating a deficit to lose body fat, while exercise is the primary tool for maintaining lean body mass during a deficit.

Chapter 5. Mechanisms of Muscle Hypertrophy

Muscle growth can occur by hypertrophy and hyperplasia. In humans performing traditional resistance training programs, increases in muscle size occur primarily via myofibrillar hypertrophy; hyperplasia appears to have little effect on growth unless pharmacological aids are employed. Although in series hypertrophy can contribute to early phase muscular gains, the majority of increases are attributable to the addition of sarcomeres in parallel. Some research suggests that expansion of sarcoplasmic fractions may account for some of the hypertrophy achieved through resistance training, although evidence on the topic remains equivocal. Moreover, hypertrophy often manifests in a non-uniform fashion, with the various heads or even portions of the same muscle experiencing different magnitudes of growth. Importantly, hypertrophy is specific to the individual, with wide-ranging responses observed between those performing the same resistance training program.

Several potential mechanisms have been proposed to drive hypertrophy. Compelling evidence indicates that mechanical tension is the primary hypertrophic driver. There is conflicting evidence as to whether metabolic stress and muscle damage play a supporting role and, if so, whether these factors promote an additive stimulus or are simply redundant given sufficient mechanical stimuli. Difficulties in controlling for confounding variables make it problematic to draw causality in human hypertrophy research designs. Moreover, other mechanisms may also be involved to some as-yet unknown extent. Although great strides have been made in our comprehension of hypertrophic mechanisms, science is a long way away from fully understanding their implications in exercise-induced growth.

Hormones play a role in muscular development. Chronic endogenous levels of testosterone and likely other anabolic hormones (i.e., IGF-1, GH, etc.) are at least permissive to the accretion of muscle mass, and at the upper and lower extremes likely influence muscular potential. Alternatively, chronically high levels of cortisol, a catabolic hormone, antagonize anabolic processes and thus can be detrimental to hypertrophic goals. Although various hormones show substantial acute increases after a resistance training bout, these transient spikes likely are not meaningfully involved in the adaptive response. Rather, acute increases in various myokines that exert their effects directly on muscle tissue in response to mechanical

stimuli appear to be the primary drivers of muscle protein accretion. Research has identified numerous myokines with potential effects on muscle development, although the specific roles of the substances and their interactions with one another remain somewhat uncertain.

From a nutritional standpoint, maximum muscle gains require an energy surplus. Relatively untrained lifters can undertake a substantial energy surplus and gain substantial lean mass with minimal increases in body fat. Alternatively, well-trained individuals require a more modest energy surplus (≤ 500 calories/day) to avoid significant increases in body fat during a mass-building phase. To maximize anabolism, protein intake for resistance-trained individuals should be in the range of 1.6 to 2.2 g/kg/day (~ 2.0 to 2.6 g/kg/day when expressed in terms of g/kg/day when expressed in terms of lean mass); as a general rule, those aspiring to achieve physique-related goals should consume at the higher end of these guidelines as there may be a potential hypertrophic benefit with little downside. While carbohydrate and dietary fat will necessarily comprise the balance of calories, there is no ideal ratio of these nutrients. It seems prudent to consume at least a moderate amount of carbohydrate (~ 3 g/kg/day) from a muscle-building standpoint given its role in fueling performance and potentially anabolic signaling. That said, a high-carbohydrate diet is not obligatory for maximizing lean mass. Alternatively, dietary fat consumption also helps to support hypertrophy. The polyunsaturated fats are essential to health, and the omega-3 fats in particular appear to aid in muscle-building. Once basic needs of carbohydrate and dietary fat are fulfilled, the additional calories making up total daily energy intake can be consumed based on preference and individual response.

Key Takeaways

- Muscle hypertrophy primarily occurs via the addition of sarcomeres in parallel. An increase of sarcomeres in series can occur in the early stages of training and perhaps with certain types of exercise protocols, but its contribution to hypertrophy is limited.
- Hypertrophy may occur via a disproportionate increase in sarcoplasmic versus contractile fractions; however, the extent to which this meaningfully contributes to hypertrophy and whether specific training protocols may influence this outcome remain to be determined.
- Hypertrophy can occur in a non-uniform fashion, with varying magnitudes of growth in proximal, middle, and distal segments of a muscle.
- Hyperplasia does not play a meaningful hypertrophic role in traditional human training protocols.
- Mechanical tension is the primary driver of hypertrophy. Evidence suggests that other factors, including metabolic stress and muscle damage, may play a role, but the extent of their involvement remains equivocal.
- Chronic levels of anabolic hormones (i.e., testosterone, IGF-1) support muscle-building processes. However, acute post-exercise hormonal elevations do not play a consequential role in the hypertrophic response to resistance training.
- An energy surplus is needed to maximize muscular gains.
- Protein intake should range between 1.6 and 2.2 g/kg to elicit an optimal anabolic response.
- Although there is not ideal carbohydrate/fat ratio for muscle-building, a modest carbohydrate intake of at least ~ 3 g/kg/day seems appropriate to fuel resistance training performance.

- The majority of dietary fat intake should come from unsaturated sources, with a particular emphasis on consuming omega-3 fatty acids.

Section 3: Nutrition Principles

Chapter 6. Macronutrients

The goal of this chapter was to provide fundamental concepts of nutrition along with evidence-based dietary recommendations for Physique Coaches and physique athletes. Since physique athletes are judged on aesthetics, nutritional recommendations are based upon the goal of supporting training and enhancing body composition (i.e., decreasing body fat and building muscle mass). Changes in body weight are primarily related to energy balance, whereby weight loss requires a calorie deficit and weight gain requires a caloric surplus. Determining a physique athlete's maintenance calories is often a good place to start.

Maintenance calories are the amount of energy intake required to maintain one's current body weight and therefore represent a calorie intake that matches total daily energy expenditure. Energy expenditure can be partitioned into distinct components based upon their relative contributions toward total daily energy expenditure: RMR, TEF, EAT, and NEAT. An estimation of maintenance calories can be calculated based upon a physique athlete's body characteristics and physical activity status using the equations and correction factors provided in this chapter. Calorie needs can then be estimated depending on the goal of weight maintenance, loss, or gain. For fat loss, slower rates of weight loss (0.5 to 1% of body mass per week) are generally preferable to attenuate the loss of lean body mass and unfavorable metabolic adaptations, and it is generally recommended to start with a 10 to 20% energy deficit (i.e., multiplying maintenance calories by 0.9 to 0.8). To maximize muscle growth in the off-season, hypercaloric conditions are ideal, yet greater caloric surpluses increase the risk of adding unwanted body fat. A modest surplus (10 to 20% above maintenance calories; ~250-500 Calories per day) seems appropriate for more advanced physique athletes who are nearing their genetic capacity to build muscle; whereas a larger caloric surplus (20 to 40% above maintenance calories; ~500-1,000 Calories per day) seems appropriate for novice-level physique athletes with greater potential for muscle growth.

Once calorie needs are established based upon the goals of the physique athlete, the next step would be to distribute the calories between protein, fat, and carbohydrate (Table: General Dietary Recommendations for Physique Athletes). Dietary protein offers 4 Calories per gram. Proteins are composed of sequences of amino acids linked together and serve several functional roles in virtually all body functions. Notably, protein is necessary to activate muscle protein synthesis and facilitates muscular growth.

TABLE: General Dietary Recommendations for Physique Athletes			
	Weight Loss (Contest Preparation)	Weight Gain (Offseason)	
		<u>Novice</u>	<u>Advanced</u>
Weekly weight change (% of body weight)	- 0.5 to 1%	+ 0.25 to 0.5%	+ 0.25
Calories	10 to 20% below maintenance needs	20 to 40% above maintenance needs (~500 to 1,000 kcal)	10 to 20% above maintenance needs (~250 to 500 kcal)
Protein	1.8 to 2.7 g/kg or 2.3 to 3.5 g/kg of lean body mass	1.6 to 2.2 g/kg	1.6 to 2.2 g/kg
Fat	15 to 30% of total calories 0.5 to 1.5 g/kg	20 to 35% of total calories 0.5 to 1.5 g/kg	20 to 35% of total calories 0.5 to 1.5 g/kg
Carbohydrates	Remaining percentage of total calories (~2-5 g/kg/day)	Remaining percentage of total calories (≥3-5 g/kg)	Remaining percentage of total calories (≥3 to 5 g/kg)
(Aragon & Schoenfeld, 2020; Helms et al., 2014a; Iraki et al., 2019; Roberts et al., 2020)			

Physique athletes require adequate dietary protein to promote muscle growth and support the maintenance of lean body mass during contest preparation. Total daily protein intake of approximately 1.6 g/kg is sufficient to maximize resistance training-induced gains in muscle mass under eucaloric or hypercaloric condition. However, under hypocaloric conditions, protein recommendations can be increased to 1.8 to 2.7 g/kg or expressed relative to lean body mass, 2.3 to 3.5 g/kg of lean body mass. Fat is the most calorie-dense macronutrient, offering 9 Calories per gram and plays a vital role in several important bodily functions. Triglycerides are the most common type of dietary fat and storage form of fat. For physique athletes, dietary fat recommendations are generally focused on achieving an adequate fat intake without compromising protein and carbohydrates intake.

During the offseason, a dietary fat intake of 20 to 35% of total calories is recommended, which would equate to 0.5 to 1.5 g/kg/day for most physique athletes. However, during periods of calorie restriction, obtaining 20 to 35% of total calories from fat may be difficult without compromising sufficient protein and carbohydrate intakes. A lower end of fat intake between 15 to 20% of total calories may be deemed appropriate during times when a higher percentage would reduce carbohydrate and protein intake below ideal ranges. Carbohydrates offer 4 Calories per gram and serve as an important energy source for the central nervous system, red blood cells, and skeletal muscle both at rest and during exercise. An optimal carbohydrate recommendation for physique athletes has not been clearly defined. One approach to determining carbohydrate intake for a physique athlete would be to first calculate the calories required to result in appropriate body weight losses/gains followed by a determination of an optimal amount of protein and fat. Subsequently, the remaining calories can be allotted to carbohydrates. A realistic carbohydrate intake range for physique athletes during contest preparation and periods of low energy intake is 2 to 5 g/kg/day.

Alcohol is a nonessential nutrient that offers 7 Calories per gram. Excessive alcohol consumption has been associated with an elevated risk of negative mental and physical health conditions, along with unfavorable outcomes that may interfere with physique-oriented goals. Alcohol intake may promote a calorie surplus, alter the hormonal milieu, impair recovery from muscle damaging exercise, reduce circulating levels of amino acids, blunt intramuscular anabolic activity, and impair sleep quality. Physique Coaches and physique athletes should monitor habitual alcohol consumption to avoid these negative consequences.

Creating a caloric deficit is imperative for fat loss to occur over time. As opposed to continuous energy restriction whereby energy intake is decreased over time in a stepwise fashion, intermittent energy restriction strategies may be employed. Using alternating periods of energy restriction with periods of greater energy intake may offset some of the common adverse physiological effects associated with contest preparation diets and attenuate unfavorable metabolic adaptations to weight loss. These strategies include incorporating a cheat meal or a cheat day, diet refeeds, and diet breaks. Although these strategies do show promise for improving fat loss efficiency, protecting lean body mass, and attenuating reductions in resting metabolic rate while improving adherence and reducing the psychological burden of continuous energy restriction, intermittent energy restriction strategies will extend the duration of the total competition preparation period and require that the physique athlete be in an even greater energy restricted state during the non-refeed days.

Several popular dieting styles can be employed to reach body composition goals. However, no dietary strategy has consistently proven to be more efficacious than others when calorie intake is controlled. Therefore, physique athletes are advised to implement such dieting strategies only if they find the dietary approach to be sustainable and enjoyable. Two popular dietary strategies among physique competitors have been the ketogenic diet and intermittent fasting. Ketogenic diets traditionally consist of a low-carbohydrate (less than 50 g per day), high-fat (60 to 80% of total calories), and moderate-to-high protein intake (1.2 to 2.0 g/kg/d). The ketogenic diet is often associated with reductions in hunger and a spontaneous decrease in calorie intake due to the elimination of carbohydrate sources. However, some evidence suggests a negative effect on intramuscular glycogen and water stores along with lean body mass maintenance, making it suboptimal for bulking and gaining muscle mass. Intermittent fasting is a dieting strategy that uses regular short-term fasts with periods of ad libitum food intake. Intermittent fasting may promote a decrease in calorie intake due to a narrower feeding window. While several studies have shown that intermittent fasting strategies are a viable and effective method for reducing caloric intake and body fat, intermittent fasting has not been demonstrated to be superior to daily caloric restriction for improving body composition or health outcomes. Intermittent fasting may also be suboptimal for muscular development given that the daily impact on muscle protein synthesis may be negatively impacted. While ketogenic diets and intermittent fasting are viable dieting strategies for physique athletes to experiment with, physique athletes should be aware of the potential downsides.

Key Takeaways

- Weight loss requires a negative caloric balance known as a calorie deficit, whereby energy expenditure exceeds energy intake. Weight gain occurs during a positive caloric balance known as a caloric surplus, whereby energy intake exceeds energy expenditure.

- Total caloric needs and recommendations for physique athletes will vary widely based upon total daily energy expenditure, body characteristics, and goals.
- Energy expenditure can be partitioned into distinct components based upon their relative contributions towards total daily energy expenditure. These components consist of resting metabolic rate (RMR), thermic effect of food (TEF), exercise activity thermogenesis (EAT), and non-exercise activity thermogenesis (NEAT).
- Once calorie needs are established based upon the goals of the physique athlete, the next step would be to distribute the calories between protein, fat, and carbohydrate.
- Total daily protein intake of approximately 1.6 g/kg is sufficient to maximize resistance training-induced gains in muscle mass under eucaloric or hypercaloric condition. Under hypocaloric conditions, protein recommendations can be increased to 1.8 to 2.7 g/kg or expressed relative to lean body mass, 2.3 to 3.5 g/kg of lean body mass.
- Dietary fat recommendations are generally focused on achieving an adequate fat intake without compromising protein and carbohydrates intake. During the offseason, it seems prudent to recommend a dietary fat intake of 20 to 35% of total calories, which would equate to 0.5 to 1.5 g/kg/day for most physique athletes. During periods of calorie restriction, a lower end of fat intake between 15 to 20% of total calories may be deemed appropriate to avoid reducing carbohydrate and protein intake below ideal ranges.
- An optimal carbohydrate recommendation for physique athletes has not been clearly defined. One approach to determining carbohydrate intake for a physique athlete would be to first calculate the calories required to result in appropriate body weight losses/gains followed by a determination of an optimal amount of protein and fat. Subsequently, the remaining calories can be allotted to carbohydrates. A realistic carbohydrate intake range for physique athletes during contest preparation and periods of low energy intake is 2 to 5 g/kg/day.
- Alcohol is a nonessential nutrient that offers 7 Calories per gram. Along with the excess “empty calorie” intake and general health risks, regular alcohol ingestion can promote other unfavorable outcomes that interfere with physique-oriented goals and muscle building.
- Despite some potential downsides to consider, intermittent dieting strategies (e.g., “cheating”, refeeds, and breaks) do show promise for improving fat loss efficiency, protecting lean body mass, and attenuating reductions in resting metabolic rate while improving adherence and reducing the psychological burden of continuous energy restriction.
- Physique athletes may experiment with a ketogenic diet in the early stages of contest preparation to appreciate if it is adherable and enjoyable; however, some evidence suggests a negative effect on maintaining lean body mass, making it suboptimal for bulking and gaining muscle mass.
- Intermittent fasting strategies are a viable and effective method for reducing caloric intake and body fat, yet they have not demonstrated to be superior to daily caloric restriction for improving body composition, cardiometabolic, inflammatory, or appetite outcomes.

Chapter 7. Micronutrients and Hydration

This chapter identified the impact of food and supplemental micronutrient intake on human physiology and metabolism. While it has been observed in previous research that physique athletes may consume less micronutrient content during physique contest preparation, it is unknown if any qualified as deficient. Similar to athletes in other sporting and recreational activities, it is recommended that physique athletes follow the RDA and AI recommendations of micronutrient intake. Additionally, there does not seem to be any performance benefit for increased micronutrient intake above recommended RDA and AI values unless a deficiency has been determined. However, because physique athletes may follow more strict dietary guidelines that limit nutrients found densely in food sources mixed with higher energy expenditure during an energy deficit to reduce body fat, the use of a multivitamin/mineral within the RDA and AI recommendations would not be considered harmful if the food choices are limited. Water intake and restrictions are utilized during the final peak week of physique contest preparation. Currently, there is no validated approach to acute water loading and restriction, nor is there any validated or recommended approach to electrolyte loading or restriction that may be optimized for health and cosmetic performance for the physique athlete. However, research identifying a water loading protocol has shown to promote fluid and weight loss with limited impact on markers of hydration during water restriction. Overall, it is recommended that Physique Coaches and physique athletes follow validated and most generalizable evidence to make decisions regarding micronutrition and water intake and restriction for physique competitions.

Key Takeaways

- Micronutrients are a class of organic (vitamins) and inorganic (minerals) nutrients in comparison to macronutrients (carbohydrates, lipids, and protein), yet are required in a much lesser daily amount.
- Vitamins are categorized naturally in two forms: water-soluble vitamins and fat-soluble vitamins. Water-soluble vitamins may be more readily lost through body fluids (urine), so they need to be ingested regularly, whereas fat-soluble vitamins may be stored and accumulate in the liver and fatty tissues (subcutaneous adipose tissue), so their consumption may need not be required daily.
- Minerals are classified into macrominerals and microminerals or trace minerals. Macrominerals are required by the body in larger amounts whereas trace minerals are required in much smaller amounts. Minerals must be consumed regularly, ranging from grams per day (g·d⁻¹; macrominerals) to milligrams per day (mg·d⁻¹; trace minerals).
- It is important to consider that the excessive intake of a micronutrient may alter or compete with the absorption and health benefits of another or increase the risk of toxicity.
- It is well known and reported that physique athletes (bodybuilders, figure, bikini, physique) generally recruit a stricter dieting practice when preparing for a competition. They may either minimize certain food intake or eliminate certain food groups together. Due to this behavior, this may warrant micronutrient supplementation in this population.
- Ergogenic and health claims associated with micronutrient intake have been refuted in healthy, non-micronutrient deficient populations. In populations that are older or

diagnosed with a deficiency, there is support that increased micronutrient intake may influence performance measures.

- The increase in fitness social media influencer marketing has made it easier to use false or misleading claims about dietary supplements to shape health and fitness consumer trends. Physique Coaches and physique athletes should use great caution and skepticism when interpreting supplement claims on social media due to a lack of accuracy and validity.
- The homeostatic maintenance of normal hydration status is important for circulatory function, biochemical reactions that drive metabolism, cellular transport of nutrients and waste products to and from tissues, and temperature regulation.
- Hypohydration has been shown to impair exercise through a range of mechanisms, including a reduction in blood plasma, impaired cardiovascular function, muscle blood flow, thermoregulatory capacity, and neuromuscular function via strength and power.
- Physique Coaches and physique athletes can determine hydration status by using simple biomarkers (urine and body weight) that are known to have accuracy limitations, yet when interpreted in the proper context, they can provide a more objective measure that may give valuable insight into hydration status.
- Recommendations for peak week fluid intake and restriction, sodium intake, and restriction are highly subjective in that there is no research to support that there is an optimal approach to water manipulation to optimize the reduction of subcutaneous fluid (“dry”) content while maximizing muscular tonicity (“full or hard”). Additionally, it is unknown how water loss due to water restriction and increased urination would influence extracellular and intracellular shifting between these compartments in a physique athlete population.
- Currently, there are no recommendations for sodium loading or restriction during peak week along with no generalizable research to support that manipulating sodium intake has any cosmetic performance benefit.
- A recommended and conservative approach to peak week Na^+ and K^+ manipulation is to simply maintain current intake as opposed to irrational adjustments based on personal or other anecdotal experiences.

Chapter 8. Supplementation

Assessing a physique athlete’s nutrition in a food-first manner is important as a Physique Coach. This chapter focused on analyzing some of the commonly recommended dietary supplements for muscle hypertrophy and fat loss, examining their efficacy for performance enhancement, benefits, and risks associated with their use. With a saturated market of dietary supplements, it can be difficult to navigate and fall prey for marketing hyperbole. Following an evidence-based approach, a decision can be made on whether a physique athlete will benefit from dietary supplementation, or whether the body of evidence supports the use and efficacy of a particular dietary supplement, hopefully removing some of the fallacies that exist within the fitness industry of supplementation being a strict requirement.

Key Takeaways

- The dietary supplement market is growing at an exponential rate and continues to gain popularity as a means to support well-being alongside sports performance.

- As a Physique Coach, it's important to assess and understand the individual needs basis for supplementation of clients. Focusing on a food-first approach should always be at the forefront before a strategy is implemented for dietary supplements.
- Understand the legalities, liabilities, and technicalities when proposing dietary supplements to clients. Under no circumstance should a prescription be made.
- While dietary supplements can be beneficial alongside a food-first approach, an understanding of the biochemistry, mechanism of action, and safety data is critical before any recommendation.
- The Journal of the International Society of Sports Nutrition (ISSN) publication "ISSN exercise & sports nutrition review update: research & recommendations" is a comprehensive resource for understanding dietary supplements reviewed and categorized based on the weight of their evidence.
- Dietary supplements can be useful for muscle building and sports performance when accompanied alongside a well-structured nutrition and exercise plan.
- Similarly, dietary supplements can be useful for fat loss, but a deficit in energy expenditure is still foremost critically important.
- Contamination of dietary supplements not only brings serious health consequences, but under the principle of strict liability, physique athletes competing in tested federations have potentially a 30% chance of taking a supplement that contains an adulterated substance not listed in the ingredients. The Tainted Products Marketed as Dietary Supplements CDER database is a useful resource to aid in lowering that risk.

Chapter 9. Performance Enhancing Drugs

Aside from dietary supplements, PEDs are another category of ergogenic aid that have unfortunately become part of a scribe of recommendations to physique athletes. As a Physique Coach and fitness professional, it is imperative that there is a vital understanding to the mechanisms of action of commonly used PEDs for muscle hypertrophy and fat loss, in order to appreciate and coach on the dangers of the substances to physique athletes. As a Physique Coach, the power of the placebo and nocebo is to be appreciated, having both positive physical, and psychological outcomes towards improved sports performance if the physique athlete has a strong psychological belief that what they are taking indeed works. Nevertheless, medical referral should be the appropriate response following the disclosure or approach towards the use of PED by any physique athlete to ensure the health of the physique athlete is adequately managed in both an acute and chronic setting.

Key Takeaways

- Performance-enhancing drugs (PEDs) are an inevitable part of bodybuilding and sport, with their use bringing serious public health discussion and stigma.
- Athletes may be tempted to use PEDs from both intrinsic (self) and extrinsic (external) sources of motivation.
- PEDs can endanger both current and future health depending on the mechanism of action, dose, and duration of use.

- As a Physique Coach, it is beyond the scope to self-diagnose or medically treat a client who wishes to partake in PED use or is already engaged in their use.
- Referral to a medical practitioner is critical, and every attempt should be made to place this emphasis to a client for regular health screenings.

Section 4: Body Aesthetics, Fitness and Nutrition Assessments

Chapter 10. Fitness and Body Aesthetics Assessments

A coach should use fitness and body aesthetic assessments with their clients to start working towards a competition preparation and to track their client's progress. This chapter covered why some of these assessments are done before the Physique Coach starts working with the client so that they can determine what areas need the most work, while other assessments are repeated on a consistent basis.

The Physique Coach will use various questionnaires, such as the PAR-Q+, to capture the client's history and relevant medical conditions. They will also need to use a physique-specific health history questionnaire to determine the client's baseline knowledge and experience level and to gather important data before creating the initial program. There are also physiological assessments, like resting heart rate, blood pressure, and blood tests, that the Physique Coach can use initially and in an ongoing fashion. Blood markers like thyroid, cortisol, testosterone, estrogen, and vitamin D will fluctuate throughout the offseason and competition prep phases, so revisiting these blood markers can be informative.

Body composition assessments are taken both initially and on a frequent basis determined by the Physique Coach. Physique athletes need to compete at a very specific level of muscularity and leanness for the division they are competing in, and these assessments will track these division-specific differences. Taking starting photos and recording circumference measurements is something the client can do initially by themselves and will continue to send the Physique Coach on a basis that the coach determines is necessary (typically, a week). Body fat measurements, whether through skinfolds or BIA, are other cost-effective ways to track progress, but may not be accessible to every client.

Assessing the client's body aesthetics, between muscularity, symmetry, and proportions during the initial assessments allows the Physique Coach to determine what areas need to improve. Male and female clients will hold body fat differently, which can make it hard to determine how much muscle needs to be added in an offseason period, but that is why a variety of these body assessments are repeated on a consistent basis.

Lastly, this chapter covered the importance of effective and professional communication strategies the Physique Coach should use during the initial consultation, whether it is conducted virtually or in person. Asking open-ended questions, being attentive, using the appropriate language, and creating a safe and non-judgmental environment are all necessary to build rapport and trust with the client. The Physique Coach must also clarify expectations for the client and make that specific to their experience level. Working with a first-time competitor

versus an experienced competitor, will not only differ in the program and protocols designed, but the expectations that are created.

Key Takeaways

- Participant health screenings consists of questionnaires to understand the history, medical conditions, and experiences of a client. These demonstrate professionalism, build rapport between coach and client, protect the Physique Coach from any unknown concerns before starting to work with the client, and create the foundation of the client's program.
- Physiological assessments, like resting heart rate, blood pressure, and blood tests, should be taken at baseline to gather information about the client's health, and be retested later during a competition prep for comparison.
- Relevant blood tests to run for physique competitors include values on cortisol, thyroid hormones, testosterone, estrogen, and vitamin D. If working with enhanced athletes, other tests, such as blood lipids, liver enzymes, and complete blood count, are useful.
- Body composition assessments should be tracked all throughout a client's offseason and competition prep period. Progress pictures, circumference measurements, and body fat tests are all validated ways to assess a client's change week to week.
- Muscularity, symmetry, and proportion are all important body aesthetic metrics the coach should evaluate during their initial assessment to create the appropriate program and develop a timeline for the client to step on stage. • The initial client consultation is the time when the coach will make their initial assessments of the client, provide constructive feedback and a framework for both the offseason and competition prep period, as well as provide space for the client to ask any questions about the process.
- Scheduling and performing the initial client consultation will differ distinctly between in-person and online settings, but with either method, the coach can gather the necessary information to get the client started.
- Coaches need to develop effective communication strategies so their clients can implement change successfully. Asking open-ended questions and providing autonomy-supported feedback are two ways to encourage behavior change in clients.
- During all points of the assessment, coaches should display positive body language, show attentiveness, use appropriate language, and provide a private, non-judgmental environment for the client.
- Clarifying expectations for the client is important and will vary between experience level. Coaches will need to tailor their message to the novice, intermediate, and advanced competitor, as well as the competitor who had a previously negative experience.

Chapter 11. Nutrition Assessment

Without effective assessments and tools to evaluate a client accurately, the Physique Coach will be unable to effectively coach and guide their clients to a successful outcome. Both assessment and evaluation allow the coach to measure and make determinations and judgements about the progression of a client's journey and the effectiveness of the program the Physique Coach and client are working on implementing together.

The assessment and evaluation period often begins at the outset of the relationship during the preparation for the first consultation and during the first consultation, but it does not stop there. They are ongoing and iterative processes that require feedback and discussion from the client and continuously improving and adapting the assessment and evaluation tools to fit the client where they are at in their journey.

There are specific considerations related to assessment and evaluation the Physique Coach should place extra emphasis on considering the type of results physique clients chase and interventions that are required to achieve those results.

Key Takeaways

- While energy balance is a primary focus for achieving many body composition goals, assessing and evaluating eating patterns to ensure a client is following healthy eating guidelines is important.
- Understanding the pros and cons of various forms of tracking food intake (e.g., diet records, 24-hour recall, food frequency questionnaires) and selecting the most appropriate one for each client at each stage of their program/journey.
- Provide education and guidance on how to effectively use mobile apps for tracking food intake as they have many pitfalls and errors that affect physique clients.
- Weighing and measuring food is one tool that physique clients may rely heavily on during the more focused parts of the physique transformation process. It is important that the Physique Coach provides guidance on best practices for this method of food logging and tracking.
- Help clients identify hunger and to self-identify physical hunger cues and emotional hunger cues as well as how to understand satiety and change dietary habits to better regulate satiety.
- Data collection, analysis, interpretation, and communication is critical to the success of the coach–client relationship and is an area that the Physique Coach should spend time developing skills around.

Section 5: Exercise Training Principles

Chapter 12. Warm-up, Recovery, and Injury Prevention

Training physique athletes requires careful consideration for not only their predispositions to musculoskeletal injuries, but also to accepted norms within the bodybuilding culture (e.g., training through pain). As such, the Physique Coach should have knowledge of the relevant conditions that may affect physique athletes as well as strategies to address postural and movement dysfunctions that may impede performance and at worst, cause injury.

This chapter introduced the importance of a warm-up within the training routine of bodybuilders and provided evidence-based support for its use alongside NASM's Corrective Exercise Continuum. Proper integration of the Corrective Exercise Continuum was discussed based on the findings of a systematic assessment. This chapter covered the pertinent thought processes that should be considered when working with physique athletes (e.g., Chris's case).

Last, the chapter finished with a discussion of evidence-based recovery approaches with relevant bodybuilding considerations. Incorporating this chapter's content into the programming of a bodybuilder will improve posture, muscle activation, and strength, and likely reduce the risk of future injury.

Key Takeaways

- A warm-up is an exercise or series of exercises which has a primary goal of preparing the body for strenuous exercise.
- Warm-ups can be successfully implemented prior to the training session of a physique athlete to increase performance, improve posture, and retrain movement dysfunctions that may predispose the bodybuilder to injury.
- A comprehensive warm-up encompasses a general component that elevates body temperature typically through aerobic exercise followed by integration of the Corrective Exercise Continuum just prior to a specific warm-up using lower intensities of major compound movements.
- The Corrective Exercise Continuum is a stepwise, systematic approach to addressing postural and movement dysfunction that incorporates four steps: inhibit, stretch, activate and integrate.
- All steps in the Corrective Exercise Routine are designed to reduce compensations and optimize performance during exercise which may lead to better hypertrophy—the primary objective of bodybuilding.
- Physique athletes will likely experience overuse injuries due to the moderate to heavy loads and high volumes of training in their programming.
- Overtraining syndrome is the result of a prolonged period of high resistance training volume (greater than a mesocycle) without adequate rest and recovery leading to negative symptoms and heightened injury risk.
- Physique Coaches should be able to recognize the major signs associated with overtraining including insomnia, tachycardia, high blood pressure, and irritability.
- Corrective exercises should not be integrated into a workout routine of a bodybuilder without an appropriate assessment.
- The assessment process should include a general intake form and a static and dynamic posture assessment including the Overhead Squat Assessment.
- The Overhead Squat Assessment is an important whole-body dynamic movement assessment that correlates observed movement dysfunctions with muscle under or overactivity and limitations in neuromuscular recruitment patterns and range of motion.
- When more than one dysfunction is present in the Overhead Squat Assessment, consideration should be given to the most relevant postural dysfunction observed to impact the training routine.
- Physique athletes are likely to experience musculoskeletal pain during their sessions and recognizing the signs and symptoms that warrant a referral to a rehabilitation provider is important.
- Recovery is a buzzword in the fitness and rehabilitation community, and in the context of physique athletes, means the ability to perform to the same degree or greater compared to a previous bout of exercise.

- The evidence at large supports adequate nutritional intake, hydration, and sleep as most important to improve recovery.
- Other approaches including cold water immersion, massage, and vibration therapy may be carefully implemented as long as they are following recommended guidelines for the more important recovery tools.

Chapter 13. Resistance and Core Training

Training physique athletes requires an advanced understanding of foundational resistance training principles, a wide array of acute variables, and intelligent use of techniques to elicit the appropriate type and dose of stimulus to maximize muscle growth and body composition outcomes. A Physique Coach must tailor a program's volume, intensity, and frequency to the individual athlete's needs. Furthermore, to maximize their muscular development, strategic exercise variations should be programmed and periodized to ensure all muscles are being trained through their full range of motion and all muscle actions are being directly trained. Although the level of muscularity required for different divisions of physique athletes varies, a comprehensive training program is not complete without proper core training. Core training is crucial to keep a physique athlete healthy and progressing in the gym by reducing their risk of injury. Moreover, a well-developed midsection can improve the physique athlete's overall aesthetics, enhance the flow of their physique, and tie in their other muscle groups like a beautiful centerpiece.

Key Takeaways

- There are seven training principles that Physique Coaches must understand: Specificity, Progressive Overload, Progression, Individuality, Variation, Reversibility, and Diminishing Returns.
- The acute training variables will be primary tools in a Physiques Coach's toolbox to organize their clients' program to maximize its effectiveness. Acute training variables include volume, intensity, frequency, exercise selection, exercise order, inter-set rest periods, and repetition tempo.
- There are many resistance training systems and modalities that Physique Coaches can use with their clients. These systems and modalities include various training splits, warm-up sets, single sets, multiple sets, pyramid sets, supersets, compound sets, drop sets, cluster sets, eccentric training, and internal and external focus.
- Training the core musculature has a multitude of benefits for a physique athlete, including the reduced risk of injury, low back pain, and improved aesthetics.
- Two techniques that can be used to engage the core musculature include the drawing-in maneuver and the abdominal vacuum.

Chapter 14. Cardiorespiratory Training and Non-Exercise Physical Activity

Exercise can take many forms and provide a myriad of health benefits, whether it's being utilized in the context of elite-level athletics or day-to-day recreational activities. Physique athletes regularly engage in vigorous, hypertrophy-focused resistance training, which associated with numerous positive impacts on health and wellbeing. However, even well-

trained lifters can obtain additive health benefits from reduced sedentary time and increased physical activity via participation in structured cardio, recreational exercise, or various forms of non-exercise physical activity.

In the offseason, physique athletes should seek to achieve minimum physical activity guidelines for health and wellness, which entails completing at least 150 weekly minutes of activity at “moderate” MET levels (3.0 to 5.9 METs), at least 75 weekly minutes of activity at “vigorous” MET levels (≥ 6.0 METs), or a combination of the two. On the surface, this may sound like a substantial time and effort investment, but many chores, recreational activities, and activities of daily living exceed a MET level of 3. For individuals who prefer a heuristic that is easier to quantify, pronounced health benefits are associated with routinely achieving daily step count totals in the range of 8,000 to 10,000 for individuals under 60 years old, and 6,000 to 8,000 for individuals aged 60 or older. Physique athletes are also likely to positively impact their health by breaking up continuous blocks of sedentary time (that is, time spent engaged in behaviors performed in a seated, reclined, or lying position with a MET value ≤ 1.5). Sedentary time can be broken up via planned bouts of traditional physical activity throughout the day, or by incorporating extremely brief exercise snacks on a regular basis. Whether it's the offseason or contest preparation period, physique athletes stand to gain considerable benefits related to blood lipids, blood pressure, chronic disease risk, and risk of all-cause mortality by achieving minimum physical activity guidelines, regularly meeting advisable step count targets, and limiting extended periods of sedentary time.

Research has shown that exercise interventions have underwhelming independent effects on fat loss, largely due to compensatory increases in hunger and energy intake, along with compensatory reductions in non-exercise components of TDEE. Nonetheless, when implemented in combination with reductions in energy intake, physique athletes can use cardio as a fat loss tool during contest preparation. It is important for physique athletes to remember that resistance training is their primary training focus, and that fat loss is primarily driven by dietary adjustments. However, cardio can be used to facilitate negative energy balance and fat loss by increasing energy expenditure, and the impact of a given exercise bout on TDEE can be estimated using the equations described in this chapter. Cardio can span a wide range of intensities, including LISS, MISS, HIIT, and SIT, and physique athletes have plenty of flexibility to schedule bouts of cardio with a modality, duration, intensity, frequency, and volume that suits their preferences and maximizes their enjoyment. Research to date has not conclusively suggested that fasted cardio is more efficacious for fat loss than cardio completed in the fasted state, nor has research conclusively suggested that any particular intensity range (LISS, MISS, HIIT, or SIT) is superior in this context.

While physique athletes enjoy tremendous flexibility with regards to cardio programming, they should be mindful of the fact that extra cardio, and high-intensity cardio in particular, introduces an additional recovery burden to the weekly exercise schedule. Furthermore, there is a possibility that cardio might interfere with muscular adaptations to resistance training and threaten lean body mass accretion or retention. The likelihood of such interference is heightened when very large weekly doses of cardio are utilized, or when cardio is routinely completed prior to (and in close proximity to) resistance training sessions. For these reasons, physique athletes who elect to use cardio as a fat loss strategy during contest preparation should incorporate it judiciously, conservatively, and progressively over the course

of a contest preparation period. The rate of progression should be dictated largely by the need to increase energy expenditure in order to sustain a calorie deficit and by the athlete's capacity to accommodate and recover from the physical demands of any high-impact cardio modalities they choose to incorporate. When utilized strategically, cardio can be an excellent tool for supporting fat loss during contest prep, and potential downsides related to excessive training loads, insufficient recovery, and the interference effect can be largely attenuated by strategically programming sessions in accordance with the FITTE-VP principle and adhering to some simple heuristics related to cardio timing, duration, frequency, intensity, and progression.

Key Takeaways

- Many physique athletes can obtain health benefits related to blood lipids, blood pressure, chronic disease risk, and risk of all-cause mortality by reducing sedentary time and increasing physical activity via participation in structured cardio, recreational exercise, or various forms of non-exercise physical activity.
- In the offseason, physique athletes should seek to achieve minimum physical activity guidelines for health and wellness. Such guidelines advocate for at least 150 weekly minutes of activity at "moderate" MET levels (3.0 to 5.9 METs), at least 75 weekly minutes of activity at "vigorous" MET levels (≥ 6.0 METs), or a combination of the two.
- Daily step count can be used as a convenient proxy for an individual's general physical activity level. Pronounced health benefits are associated with routinely achieving daily step count totals in the range of 8,000 to 10,000 for individuals under 60 years old, and 6,000 to 8,000 for individuals aged 60 or older.
- Cardio can be used to facilitate negative energy balance and fat loss by increasing energy expenditure, and the impact of a given exercise bout on TDEE can be estimated using the equations described in this chapter. However, exercise interventions have underwhelming independent effects on fat loss, largely due to compensatory increases in hunger and energy intake, along with compensatory reductions in non-exercise components of TDEE.
- Physique athletes can successfully incorporate cardio with a wide range of modalities, durations, intensities, frequencies, and volume levels that suit their preferences and maximize their enjoyment. Common approaches to structured cardio include LISS, MISS, HIIT, and SIT, which each have unique intensity and duration characteristics.
- In some circumstances, cardio can potentially interfere with muscular adaptations to resistance training and threaten lean body mass accretion or retention. The likelihood of such interference is heightened when very large weekly doses of cardio are utilized, or when cardio is routinely completed in close proximity to resistance training sessions.
- Ideally, LISS sessions should be ≤ 60 minutes each, MISS sessions should be ≤ 45 minutes each, and both HIIT and SIT sessions should be limited to a total duration of ≤ 25 minutes each.
- Physique athletes should, when possible, limit their total number of weekly cardio sessions to no more than seven. In addition, it would be advisable to limit higher-intensity cardio bouts (including HIIT and SIT) to no more than three per week.
- When possible, cardio sessions should occur on non-lifting days. When this is not feasible, cardio should either occur in a separate session (≥ 6 hours apart from the lifting session), or immediately after resistance training within the same exercise session. For high-intensity cardio bouts (including HIIT and SIT), it may be advantageous to allow at least 24 to 48

hours of recovery before the next resistance training session targeting the same musculature.

- Fat loss is primarily driven by dietary adjustments during contest preparation, but cardio can also be used as an effective adjunct fat loss strategy. Physique athletes who use cardio to facilitate fat loss during contest preparation should incorporate it judiciously, conservatively, and progressively over the course of a contest preparation period.

Section 6: Programming and Coaching Principles

Chapter 15. Coaching Principles

Developing coaching principles that motivate and support the client's growth and behavior change is the foundation for creating a successful coach-client relationship. This chapter covered coaching principles from both a broad view and a granular approach to give Physique Coaches the tools they need to build rapport, increase compliance, and create behavior change for their clients. These principles can be applied for the client in both the offseason and contest preparation phase and are applicable for the Physique Coach whether they are working in person or virtually with their clients.

There are a number of validated goal-setting methods the Physique Coach can use with their clients. Understanding which method is most appropriate for the client will be determined by their experience level and how they are responding weekly. Learning to effectively communicate new ideas and information to the client is a necessity. Without proper communication, the client will be stuck in a phase of confusion and noncompliance. Physique Coaches can use techniques like motivational interviewing and some of its subcomponents, like asking open-ended questions and providing affirmation and reflections, to work through ambivalence with clients and create lasting behavior change.

The barriers to behavior change that physique clients experience is quite large, and often unique to this demographic. The typical barriers to behavior change, like time management, negative social influences, and all-or-nothing thinking, affect the physique client in a much larger sense given the extreme nature of the sport. Identifying how to set realistic goals and working through periods of low motivation are also imperative for the Physique Coach to relay to their client.

Personal motivation strategies are also necessary for the client to develop as the Physique Coach can develop the plan but cannot execute it. Self-monitoring behaviors, developing self-efficacy, and using positive self-talk are all strategies the Physique Coach can guide clients to create and use on their own.

Developing values is what coaching principles centers around. Without having an ethos to put the client first, the Physique Coach will struggle to ever make impactful change in their clients. Following principles centered on the importance of the offseason, foundational habits, long contest preparation periods, making sure the client is doing this for the right reason, and successfully transitioning clients through the post-competition phase are all essential for both the Physique Coach and client. Without these values anchoring the decisions, the client will

likely have a bad experience or a shortened career, and the Physique Coach will not be operating from a space that puts the clients physical and mental health first.

Key Takeaways

- Setting specific, hard goals produces greater task performance. The physique athlete will need to set specific and distinct, goals for both the offseason and competition prep.
- While physique competitions might seem as if they are exclusively extrinsically motivated, encouraging intrinsic motivation makes the process more meaningful for the client and can allow for a more successful competitive career.
- While motivational interviewing is typically framed in a consult-based setting, the Physique Coach can incorporate aspects of motivational interviewing during the initial consult and the check-in process to dive deeper into motivation and struggles the client is experiencing.
- Delivering communication expectations during the initial consultation is essential for the Physique Coach and client to be on the same page about what appropriate communication looks like for both parties during the coaching process.
- A client may experience barriers to behavior change in terms of their own ambivalence or dips in motivation, or they might be influenced more heavily by negative social influences, like an unsupportive network or demeaning comments.
- Creating strategies that support a mindset of flexible restraint over rigid restraint for clients during a non-dieting period will allow them to navigate the rigors of competition prep with better tools.
- When the Physique Coach encourages open dialogue and uses effective communication strategies, clients are more likely to be open with the challenges they are facing and negative self-talk they might be experiencing. It's through this open dialogue that the Physique Coach can help facilitate reframing the negative self-talk.
- Physique competitors have unique physiological and psychological demands placed on them, so coaches need to develop specific values and operate off an ethos that puts their clients' best interests first.
- Encouraging a long competition prep will allow the client to have more gradual fat loss, which will preserve muscle mass and training performance, and ensure the client is lean enough for their respective division.
- Developing a plan that allows the client to successfully transition from the competition prep and stabilize post-show is often overlooked but critical for the client's success in subsequent competition preps.

Chapter 16. Exercise Programming

Designing exercise programs for physique athletes can be a complicated process because there are many variables to consider such as the individual's goals, assessment results, and competition schedule. In addition, exercise programs need individuality and uniqueness to increase its effectiveness and adherence to the program. Consequently, it is vital that Physique Coaches design exercise programs that are based on the client's goals, needs, and abilities. Moreover, the Physique Coaches should adopt an integrated, multicomponent approach to program design that includes flexibility, cardiorespiratory, core, and resistance training to maximize training adaptations. This is the basis for NASM's OPT model.

The OPT model consists of three levels (stabilization, strength, and power) subdivided into five phases. The OPT model is an evidence-based model, meaning it has been both tested in a laboratory setting and successfully implemented with real-life clients and athletes to achieve their fitness, performance, and physique goals. The OPT model is useful for helping clients reduce their bodyfat and increase lean muscle mass to optimize body recomposition goals.

This chapter also provided client case studies to help the Physique Coach learn how to initiate, manage, and integrate fitness plans for three different clients. Each client presented a unique set of circumstances regarding their body composition goals, including their initial starting weight and overall health, exercise history, exercise preferences, and special considerations. This chapter aimed to provide straightforward, easy-to-follow examples for how a Physique Coach can set up, implement, and evaluate an integrated program for their clients.

Key Takeaways

- Exercise programs need individuality, based on the client's assessment results, to maximize their effectiveness.
- Exercise programs must consider many variables, such as teaching proper movement patterns, improving mobility and stability, enhancing endurance, and reducing the likelihood of injury.
- A training plan determines the forms of training to be used, how long it will take, how often it will change, and what specific exercises will be performed.
- Periodization is a systematic approach to program design that varies the amount and type of stress placed on the body to produce a physical adaptation and reduce the likelihood of overtraining and injury.
- The OPT model consists of three levels: stabilization, strength, and power.
- The OPT model includes five unique phases of training: Phase 1: Stabilization Endurance Training, Phase 2: Strength Endurance Training, Phase 3: Muscular Development Training, Phase 4: Maximal Strength Training, and Phase 5: Power Training.
- The OPT workout template is divided into six parts: Warm-Up, Activation, Skill Development, Resistance Training, Client's Choice, and Cool-Down. Cardiorespiratory training can be integrated into any section of the OPT template.
- Phase 1, Stabilization Endurance Training, teaches clients how to perform proper movement patterns, including pushing, pulling, pressing, squatting, hip hinging, and overall movement competency.
- Once clients display adequate movement patterns, Phase 1 programs are progressed by placing a greater emphasis on enhancing proprioception, balance, and postural control.
- Phase 2, Strength Endurance Training, is a hybrid form of training that involves the use of superset training in which a strength-focused exercise is immediately followed by a stabilization-focused exercise with similar biomechanical motions.
- Phase 3, Muscular Development Training, is designed to enhance muscle hypertrophy and body recomposition goals using a high volume of strength-focused exercises.
- Phase 4, Maximal Strength Training, requires the inclusion of heavy resistance training exercises to increase muscular strength.
- Phase 5, Power Training, uses superset techniques to increase the rate of force production. These superset techniques include performing a heavy resistance training exercise

immediately followed by an explosive power-based exercise with similar biomechanical motions.

- The OPT model is an exercise program model that uses various forms of periodization to help physique athletes increase muscle mass and reduce body fat.

Chapter 17. Nutrition Programming

Being able to appropriately program nutritional recommendations for clients using an evidence-based approach is a crucial component of a successful Physique Coach. This chapter focused on the core nutrition-related skills that a Physique Coach needs to appropriately program nutritional recommendations. These skills include (a) understanding the science behind the concepts of energy balance and weight change, (b) determining appropriate starting nutritional recommendations (e.g., calorie and macronutrient targets) to the client, (c) tailoring the recommendations based on the client's progress, needs, goals, and personal preferences, (d) being knowledgeable about different nutrient timing strategies that may help their client increase their chance of success, and (e) being able to monitor a client's progress over time and provide appropriate and professional feedback to the client via formal and informal communication.

Key Takeaways

- Energy balance is the most important variable determining whether an individual gains or loses weight. Therefore, recommending an appropriate number of calories is an integral skill that coaches should feel comfortable executing.
- An individual's energy requirements can be estimated using predictive equations that estimate their BMR. Their BMI can then be multiplied by an appropriate activity factor to estimate their maintenance calorie requirements. Maintenance calories can then be adjusted depending on the client's goal (e.g., weight loss).
- Aside from determining appropriate caloric targets, the Physique Coach should be comfortable recommending appropriate macronutrient targets. The appropriate quantity of each macronutrient recommended should be based on general guidelines as well as the client's individual preferences.
- The Physique Coach should provide additional nutritional strategies that may be helpful to the client. These include different nutrient timing strategies such as pre-workout nutrition tips, diet breaks, and nutrient timing.
- Providing starting recommendations is only one aspect of the Physique Coach's job. Monitoring the client's progress over time is a much larger component of the Physique Coach's role.
- The Physique Coach should have tools in place to track the client's progress over time and ensure that they stay on target to achieve their goals.
- Overall, the role of the Physique Coach is to guide the athlete to success, the same way that an NBA coach guides their team to winning a championship. The Physique Coach should be able to collect information from the client, analyze the information to formulate an effective plan, and be able to provide feedback in areas where the client can improve in order to increase their chances of success.

Chapter 18. Peaking for Competition or Personal Goals

The Physique Coach has the opportunity during peak week to fine tune the athlete's appearance before a bodybuilding competition, a photoshoot, or both. It's important to recognize that these appearances represent the culmination of months and perhaps even years of work preceding peak week, and that peak-week adjustments cannot make up for consistent and effortful training and posing, adherence to a pre-contest diet of appropriate duration, and the other preparatory efforts that must precede peak week.

A Model Strategy of peak week has been presented as series of coordinated strategies that could be employed to ensure the athlete presents a peaked physique on the day of the competition or photoshoot. However, these methods are entirely contextual and in some cases may not need to be employed at all. Most importantly, if an athlete is planning on employing peak week manipulations, a trial run performed two to four weeks before competing is encouraged to evaluate these methods in the event they need to be adjusted or eliminated during peak week itself.

This chapter also provided an orientation to physique competition, which takes on many forms worldwide, including events run by both drug-tested and non-drug-tested organizations, offering a wide range of divisions, each with different posing and attire requirements. After deciding on a competitive organization and competition, preparing to take the stage best begins many months ahead by learning posing, practicing posing, preparing the skin for spray tanning, tanning with sunbeds or natural sunlight, choosing appropriate attire, jewelry, and costumes per divisions rules and regulations, and attending to the practical matters of peak week by making reservations for travel and accommodations, setting up make-up, hair, and tanning appointments, and ensuring the athlete's food and fluids will be available either by preparation beforehand and/or ensuring these items can be purchased or prepared onsite. By addressing the logistics of peak week in advance and including at least one trial run before stepping on stage, peak-week stress levels can be greatly reduced, which can improve the on-stage look as well as leave the athlete free to enjoy the experience of competition.

Key Takeaways

- The goal of peak week is to optimize the look of the athlete's physique on the day of their event. For bodybuilding and divisions where there is a premium on muscularity, this would typically entail measures to ensure muscle fullness (e.g., via carbohydrate loading) as well as a dry appearance (possibly by using strategies to optimize body water in the event of water retention due to carbohydrate loading, stress, etc.).
- For the sake of illustrating how carbohydrate loading and drying out might be approached, a Model Strategy is presented here containing an integrated strategic approach to peak week. The Physique Coach would make use of the components of the Model Strategy only as needed, and not simply for the sake of having a peak week protocol. The best peak week protocol may entail doing nothing other than continuing the diet, training, and recovery strategies of the preceding weeks.
- In most cases, however, peak-week resistance training and cardiovascular training would be reduced during the days before the competition to ensure muscle fuel stores (especially glycogen) can be replenished and supercompensated to lend a full appearance to the musculature.

- The peak-week diet is generally not geared toward fat loss, but rather is coordinated with training to ensure muscle fullness and, when needed, to encourage water loss during the day before or day of the event to ensure a dry, but not flat, appearance.
- Minor adjustments can be made on the day of competition to fine tune fullness and dryness, but these are typically minor adjustments and should not be relied upon to remedy a flat or watery physique. Last-minute efforts in the hours before going onstage or doing a photoshoot often backfire.
- Bodybuilding organizations are found around the world, offering both drug-tested and non-drug-tested competitions with a large array of competitive divisions, ranging from traditional bodybuilding to physique model contests. The physique athlete and Physique Coach should consider which organization or competition fits the athlete's needs and goals, and recognize that any movement up the competitive ranks within one organization may not be recognized by another.
- In particular, the Physique Coach and athlete should be aware that posing requirements vary among different organizations, as do the rules and requirements regarding apparel and jewelry. Physique Coaches should encourage regular posing during contest preparation long before peak week, and even during the preceding offseason, especially for first-time competitors.
- The logistical preparations for competition (skin care, tanning, travel and lodging accommodations, hair and make-up) should be undertaken long before peak week so that the athlete experiences minimal stress and can enjoy the competitive experience during peak week.

Chapter 19. Physique Transformation Sustainability

Transitioning from a competition preparation to a post-competition phase is key to long-term health while ensuring clients have a good experience throughout the process. After a competition, clients may struggle with hyperphagia, hormonal dysregulation, body image issues, and using nutrition periodization (e.g., reverse diet, refeeds) to return their body to a sustainable place. These changes can last for weeks to months post-competition, which can induce periods of binge eating or overeating that may translate to bodyfat overshooting (Jackman et al., 2008; Trexler et al., 2017; Spendlove et al., 2015). These tendencies to overeat are driven by hedonic responses to food that were restricted during contest preparation (Helms et al., 2019). Fat regain, preferable in a controlled manner, is necessary for optimal health since persistent low bodyfat levels can interfere with the recovery, leading to further physiological disruptions (Trexler et al., 2014; Roberts et al., 2020).

Post-competition pitfalls can make an offseason difficult for a client. Some common issues include trying to stay too lean, binge eating, body dysmorphia, bodyfat overshooting, having unrealistic expectations, and/or not taking enough time off between competition seasons. Some physique athletes may only compete once, so avoiding these pitfalls will help them have a positive experience with competing. Likewise, some clients may not want to compete at all, but have major body transformation goals and could suffer from the same

common pitfalls. Ultimately, coaches are encouraged to return behaviors to pre-competition norms to create a sustainable environment for a physique athlete, and this will likely require behavior monitoring such as the use of POMS or other questionnaires.

The goal post-competition is to take a client from a negative to positive energy balance until a specific weight goal or sustainable physique is reached. The amount of weight gain will vary for each client, and the diet plan in this phase should ensure a positive energy balance, encouragement of different foods into the diet, and attempt to increase bodyweight in a controlled fashion to a sustainable level. Planned refeeds and/or diet breaks can be useful tool, but cheat meals and cheat days should be discouraged due to their association with binge eating (Roberts et al., 2020). As for exercise, during preparation, physique athletes will likely see a reduction in training volume, driven by fatigue and lower body mass. Post-competition, it is important to adjust their training protocols to something that they enjoy and aligns to their overall goals while preventing injury or overtraining. Therefore, Physique Coaches and athletes should initially decrease cardiovascular training in the first few weeks post-competition. After full physical and mental recovery, which can take two to six months, cardiovascular training can be increased as needed to achieve any new goals a physique athlete may have.

Key Takeaways

- Transitioning from competition preparation to a post-competition phase and beyond is key to long-term health and ensuring clients have a good experience.
- Even if losing weight at an optimal rate, contest preparation often causes hormonal dysregulation, loss of FFM, and suppression of energy expenditure via non-exercise activity thermogenesis due to the low levels of bodyfat needed to be competitive.
- Common post-competition pitfalls can make an offseason hard for a client. Some common issues include trying to stay too lean, binge eating, body dysmorphia, bodyfat overshooting, or having unrealistic expectations and not taking enough time off between competition seasons.
- Measures of body composition should not be used in the first few weeks post-competition, due to potential negative consequences on mindset. Realistically, a recovery period after a competition season is needed to return the body to baseline prior to starting another season, which could take two to six months or longer.
- While it is recommended for protein to stay elevated post-competition for satiety reasons, coaches may give more flexibility with carbohydrates and fats to be interchanged to expand food choices and reintroduce diversity.
- Coaches should use methods to track mood changes, such as POMS, to ensure behavior returns to normal post-competition.
- There are several ways to help a physique athlete transition from in-season to offseason diet, such as using a reverse diet or refeeds, and the primary need is for them to transition to a negative to positive energy balance.
- For a physique athlete to successfully transition after competition, they will need to restore most, if not all the bodyfat they lost, in addition to returning hormones such as ghrelin, thyroid hormones, testosterone, and cortisol to basal levels.

- Transitioning to a maintenance plan should involve a large reduction in bodyweight tracking and body checking via mirror or videos to help a physique athlete transition their mindset out of competition preparation and into a sustainable psychological state.
- After the initial post-show adjustments in exercise, there is a need to transition to a long-term hypertrophy-based training program, while simultaneously monitoring the physique athlete to avoid injury and overtraining.
- To maximize hypertrophy, 10 to 20 sets per muscle group per week, distributed across two sessions (or more), using two minutes between sets for multi-joint exercises and 60 to 90 seconds for single-joint exercises with exercises taken close to failure are needed. Furthermore, programs should include a variety of exercises that work muscles in different angles with a wide range of loading zones.
- There should be an initial decrease in cardiovascular training to as few sessions as possible in first few weeks post-competition. NEAT should also be allowed to recover to pre-competition levels. After full physical and mental recovery, cardiovascular training and NEAT can be increased as needed to achieve any new goals a physique athlete may have.